

metric current distribution along each dipole element. Excitation of an unbalanced mode on an LP antenna would be undesirable, as it would cause a reactive impedance mismatch at the input terminals, whether or not the unbalanced currents affected the radiation pattern.

To avoid strong coupling of the balanced and unbalanced modes on an LP antenna, one should utilize a structure on which the individual radiators, such as dipoles or the teeth of a zig-zag, are driven in a balanced mode by the quasi-TEM wave which propagates down the antenna from its apex toward its base. Then the balanced and unbalanced modes are not intrinsically coupled on the structure, and self-balancing is easily accomplished by feeding from a coaxial line whose outer conductor is made coincident with one of the arms of the LP antenna. The common types of successful LP antennas all satisfy this constraint. On the

other hand, the coaxially driven dipole array which we have studied in these experiments would be a poor prototype for an LP antenna, as excitation of the unbalanced mode on the radiators is not avoided, and mode splitting is the consequence.

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Computer-Aided Radome Analysis

DEMETRIUS T. PARIS, SENIOR MEMBER, IEEE

Abstract—A procedure is presented for predicting by computer the radiation pattern of an antenna in the presence of a radome. The case of a horn antenna radiating through an actual aircraft radome is treated in detail, and a comparison is made between computed and measured antenna patterns. Good agreement between theory and experiment is noted.

INTRODUCTION

A RADOME is defined [1] as "an enclosure for protecting an antenna from the harmful effects of its physical environment, generally intended to leave the electrical performance of the antenna unaffected." In practice, radome effects can be observed experimentally because the basic electrical properties of the radome wall differ markedly from those of air.

The procedures for analysis, design, and testing of radomes are well known and thoroughly documented [2]-[6]. This paper presents a technique in which the computer is used to predict radome effects on antenna patterns by integrating first over the radiating antenna to determine the field incident on the inner surface of the radome, and by integrating next over the curved outer surface to obtain the far field. The procedure is basically straightforward and can be discussed in terms of three general phases.

1) At a point on the inner surface of the radome wall, the near-field components of the radiated field are calculated using a previously reported technique [7].

At the same point the unit surface normal is calculated using radome loft data. The direction of incidence is also established, and the field is decomposed into two parts: one part which is polarized perpendicular to the plane of incidence, and a second part is polarized parallel to the plane of incidence.

2) Next, the complex transmission coefficients are computed for both perpendicular and parallel incidence. These are subsequently used to calculate the components of the field at a point on the outer surface of the radome, exactly opposite the point on the inner surface along the radome normal. The perpendicular and parallel components of the transmitted field are then recombined and the net tangential field is formed on the exterior surface of the radome wall.

3) The aperture field method is then applied to predict the radiated field from the system of secondary sources thus formed on the radome surface.

Although this approach appears quite simple, the details are numerous and deserve special consideration.

FORMULATION OF THE PROBLEM

To fix ideas, consider an X-band gain-standard horn radiating through a generally asymmetric radome, as shown in Fig. 1. A point in space can be specified either in

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The author is with the School of Electrical Engineering, Georgia Institute of Technology, Atlanta, Ga.

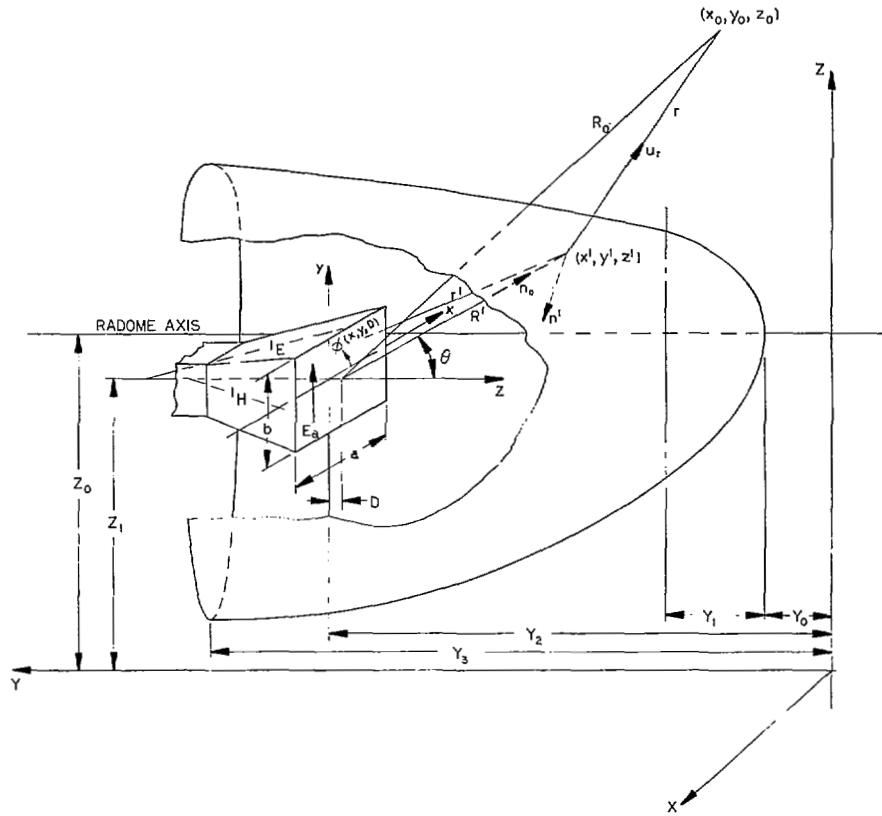


Fig. 1. Radome and horn geometry.

terms of what will be referred to as the *radome coordinates* (X, Y, Z) , or in terms of what will be called *antenna coordinates* (x, y, z) . The radome axis is contained in the YZ plane and is parallel to the Y axis. On the other hand, the xz plane is parallel to the XY plane, and the y axis is contained in the YZ plane. The horn antenna is free to rotate about the y axis through some angle Ω , as shown in Fig. 2.

The horn dimensions a and b give the size of its aperture, and l_E and l_H define the flare. Although the radome wall can be arbitrary, in this case an A -sandwich is considered which, as shown in Fig. 3, consists of two high-dielectric-constant layers separated by a low-dielectric-constant core; two external, thin layers—a rain erosion layer and an antistatic layer—complete the makeup of the five-layer wall. The thickness of each layer, as well as the dielectric constant and loss tangent, are given in Table I. The permeability of each layer is assumed to be that of free space.

A cross-sectional view of the radome through $Y = Y_0 + Y_1$ is shown in Fig. 4. Inspection of this figure shows clearly that symmetry exists only about the YZ plane. The centerline Z_0 is linearly dependent upon Y_1 , and Z_L, Z_H, R_1, R_2, R_L are all related to the same variable Y_1 according to equations of the general form

$$PY_1 + Q + (RY_1^2 + SY_1 + T)^{1/2} \quad (1)$$

where P, Q, R, S, T are all constants. It is quite obvious, of course, that the various circular segments in Fig. 4 can be expressed by equations of the form

$$F(X, Y, Z) = 0. \quad (2)$$

Specifically, when $Z_L \leq Z \leq Z_0$,

$$F(X, Y, Z) = X^2 + (Z - Z_0)^2 - R_L^2 = 0. \quad (3)$$

The following numerical tabulation completes the formulation of the problem and gives some idea on the physical size of the antenna-radome combination:

$a = 7.654$ inches	$Y_2 = 107.75$ inches
$b = 5.669$ inches	$Y_3 = 115$ inches
$l_H = 13.484$ inches	$D = 1.75$ inches
$l_E = 12.598$ inches	$Z_0 - Z_1 = 3.61$ inches
	$R_L = 16.23$ inches (at $Y = Y_3$).

The frequency of operation is 9.375 GHz.

On the basis of this information the most important aspects of the computational procedure can now be discussed. Time variations of the form $\exp(j\omega t)$ are assumed throughout this discussion.

INCIDENCE

Let Ω (see Fig. 2) be fixed at some desired value, and consider Fig. 1. At any point (x', y', z') on the inner surface of the radome wall, the generally complex x, y , and z components of the electric and magnetic field intensity vectors are first calculated using the procedure outlined in reference [7]. To meet the limitations of that procedure it is necessary that $r' \geq 1.59\lambda$ for any combination of points (x', y', z') and (x, y, D) . Additionally, it must be assumed that the tangential components of $\mathbf{E}$ and $\mathbf{H}$ vanish on the

TABLE I
DEFINITION OF THE RADOME WALL

Layer	i	κ_i	$\tan \delta_i$	d_i (inch)
Air	1	1	0	
Skin	2	4.30	0.015	0.030
Core	3	1.15	0.005	0.264
Skin	4	4.30	0.015	0.030
Rain erosion	5	3.15	0.035	0.008
Antistatic	6	7.20	0.270	0.002
Air	7	1	0	

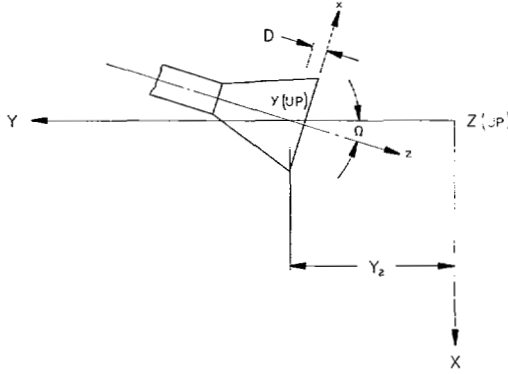


Fig. 2. The rotation of the horn antenna relative to the radome axis.

remaining portion of the closed surface which separates the source from the source-free region. A total of 12 double integrations are then made, giving both the real and the imaginary parts of each of the three spatial components of $\mathbf{E}$ and $\mathbf{H}$. In the particular case reported here, the Gauss quadrature ten-point method of integration was employed.

At the same point (x', y', z') the unit normal vector $\mathbf{n}'$ is next calculated using the formula

$$\mathbf{n}' = -(\nabla F / |\nabla F|) \quad (4)$$

where the function F is given by the left side of (2). Since the radome loft is defined in terms of the radome coordinates, the components of $\mathbf{n}'$, computed on the basis of (4), are in terms of the radome coordinates (X, Y, Z) . Ultimately, it is necessary to relate every quantity of interest to one and the same system of coordinates (preferably the antenna coordinates), using a self-consistent system of units. The following transformations are necessary for this purpose:

$$\begin{aligned} x' &= (Y_2 - Y) \sin \Omega - X \cos \Omega \\ y' &= Z - Z_1 \end{aligned} \quad (5)$$

$$z' = (Y_2 - Y) \cos \Omega + X \sin \Omega$$

and

$$\begin{aligned} A_x &= -A_Y \sin \Omega - A_X \cos \Omega \\ A_y &= A_Z \\ A_z &= A_X \sin \Omega - A_Y \cos \Omega \end{aligned} \quad (6)$$

where A_x, A_y, A_z are the components of any vector $\mathbf{A}$ measured in the (x, y, z) system of coordinates.

The calculation of the unit normal vector $\mathbf{n}'$ is required in order to establish the plane of incidence at (x', y', z') ,

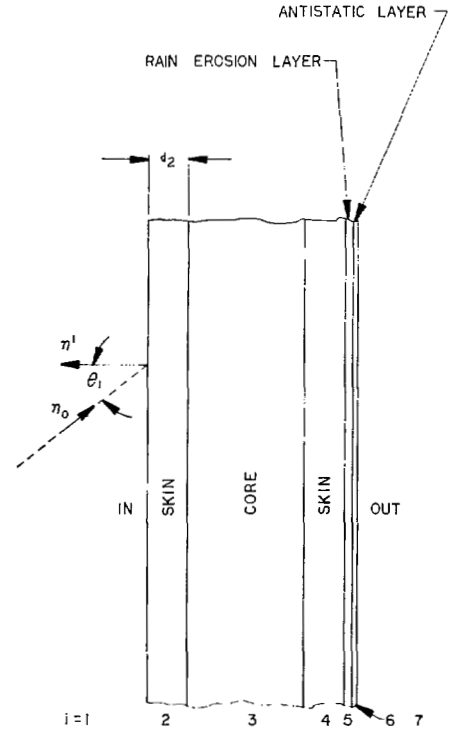


Fig. 3. Radome wall.

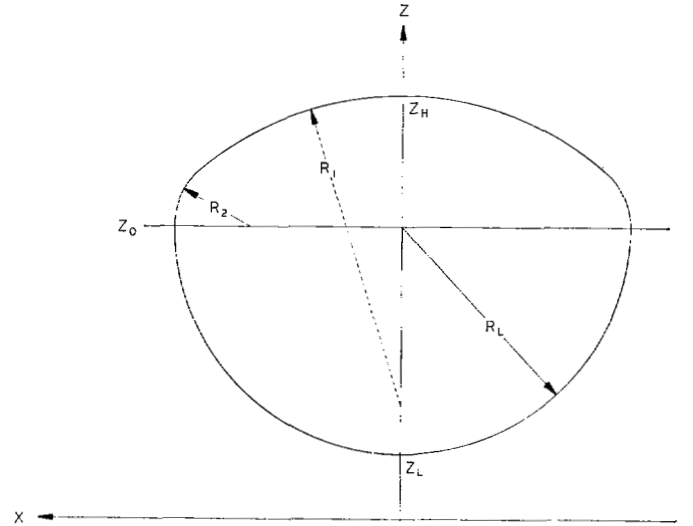


Fig. 4. Radome cross section through $Y = Y_0 + Y_1$.

since in general the incident wave will have both a perpendicular and a parallel component and since, as it is shown later, the complex transmission coefficients for perpendicular and parallel incidence are generally different for the same angle of incidence. To this end, the direction of incidence must first be established.

An obvious choice is to assume that the unit vector $\mathbf{n}_0$ establishes this direction. This is certainly true in the far field, but this choice is subject to question at points near the radiating horn. In fact, studies of the near-field pattern show that strictly speaking the incident wave is not everywhere a plane wave. For example, the relative amplitudes and phase angles of the electric field intensity components,

for $R = 0.3$ meter and $\theta = 30^\circ$, are:

$\phi = 0^\circ$	$\phi = 90^\circ$
$E_R = 0.108 \times 10^{-2}/107.5^\circ$	$E_R = 0.119 \times 10^5/97^\circ$
$E_\theta = 0.283 \times 10^{-3}/-64.1^\circ$	$E_\theta = 0.690 \times 10^5/10.4^\circ$
$E_\phi = 0.512 \times 10^5/-13.6^\circ$	$E_\phi = 0.785 \times 10^1/10.2^\circ$

where R, θ, ϕ are spherical coordinates as shown in Fig. 1. It is assumed for purposes of this calculation, that the aperture is polarized in the y direction and that, as discussed in [7], the aperture field is

$$E_a = E_0 \cos(\pi x/a) \exp[-jk(x^2/2l_H + y^2/2l_E)] \quad (7)$$

where $k = 2\pi/\lambda$ and λ is the wavelength. (The magnetic field across the radiating aperture is polarized in the negative x direction and is related to E_a according to $H_a = E_a/\eta_0$, where η_0 is the intrinsic impedance of free space.) Clearly then, while the field is virtually TEM for $\phi = 0^\circ$, it does contain a radial component for $\phi = 90^\circ$ of the same order of magnitude as the transverse component. Moreover, the phase angles of E_R and E_ϕ differ appreciably for $\phi = 90^\circ$.

The second obvious choice is to assume that the direction of real power flow can serve as a physically meaningful basis for establishing the direction of incidence at (x', y', z') . It is a known basic fact that, for sinusoidal time variations, the formula

$$\mathbf{S} = \frac{1}{2} \text{Re} [\mathbf{E} \times \mathbf{H}^*] + \frac{1}{2} \text{Re} [\mathbf{E} \times \mathbf{H} \exp(j2\omega t)] \quad (8)$$

expresses the instantaneous power flow per unit area at a point. This time-domain representation of the Poynting vector is seen to contain a term on the right which is independent of time and a second term which varies at twice the fundamental frequency of the wave. Therefore, the time-average value of $\mathbf{S}$ is not only independent of time but is also fixed in space direction. This direction can be specified by a unit vector which can be calculated easily upon dividing the vector magnitude of $\frac{1}{2} \text{Re} [\mathbf{E} \times \mathbf{H}^*]$ into its own spatial components.

Both approaches to this problem were undertaken in this investigation; although the direction of incidence was different depending on whether the first or the second method was employed, in the final analysis the far-field pattern calculated through the radome remained virtually the same. As a result, the geometrical ray theory approach (namely $\mathbf{n}_0$) was used for most calculations in this investigation. More precisely, the incidence vector was calculated according to the formula

$$\mathbf{n}_0 = (1/R')(x'a_x + y'a_y + z'a_z) \quad (9)$$

where

$$R' = [x'^2 + y'^2 + (z' - D)^2]^{1/2} \quad (10)$$

is the distance from the point of observation to the center of the mouth of the horn.

Since $\mathbf{n}_0$ and $\mathbf{n}'$ together define the plane of incidence, it is convenient to define a new unit vector

$$\mathbf{t} = t_x \mathbf{a}_x + t_y \mathbf{a}_y + t_z \mathbf{a}_z = (\mathbf{n}_0 \times \mathbf{n}') / (|\mathbf{n}_0 \times \mathbf{n}'|) \quad (11)$$

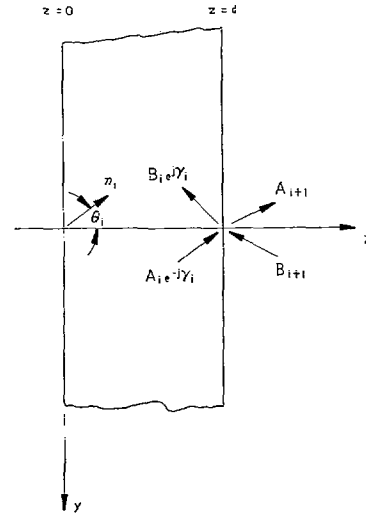


Fig. 5. The i th layer of the radome wall.

which is perpendicular to that plane and which serves as a basis for decomposing the incident field into perpendicular and parallel components. More precisely, $(\mathbf{t} \cdot \mathbf{E})\mathbf{t}$ and $(\mathbf{t} \cdot \mathbf{H})\mathbf{t}$ are the perpendicular components of the incident field at (x', y', z') , provided $\mathbf{t}$ is expressed in the antenna system of coordinates. It is obvious that $(\mathbf{t} \cdot \mathbf{E})\mathbf{t}$ and $(\mathbf{t} \cdot \mathbf{H})\mathbf{t}$ are also parallel to the radome surface at (x', y', z') , and that therefore, once transmitted through the radome wall, they can be used to compute the tangential field on the outer surface and ultimately the pattern itself. Proper transmission formulas for this purpose are developed next.

It should be noted that all components of the incident field arising from secondary reflections from other points on the radome surface are neglected in a first approximation.

TRANSMISSION

Assuming that, for any point (x', y', z') on the radome surface, the radius of curvature is large compared with a wavelength, the theory of plane waves, as it relates particularly to reflection and refraction at a plane interface [8], can be used to derive in this case a set of four recursion formulas, each of which is valid for an arbitrary number of radome wall layers, and for arbitrary angles of incidence.

Since either the electric field intensity vector (perpendicular incidence), or the magnetic field intensity vector (parallel incidence) is parallel to all the plane interfaces, four scalar quantities can be introduced by means of which the four recursion formulas can then be developed. With reference to Fig. 5, these are

- A_i : tangential $\mathbf{E}$ at left interface of i th layer, traveling left to right,
- B_i : tangential $\mathbf{E}$ at left interface of i th layer, traveling right to left,
- F_i : tangential $\mathbf{H}$ at left interface of i th layer, traveling left to right,
- G_i : tangential $\mathbf{H}$ at left interface of i th layer, traveling right to left.

At each interface, four field components are generally present. As shown in Fig. 5, at the right interface of the i th layer two waves are traveling left to right and two waves are traveling right to left. The exponentials $\exp(\pm j\gamma_i)$ describing these waves take into account both the attenuation and phase shift of the waves as they go through each layer. More precisely, the wave traveling left to right is represented by an expression of the form

$$A_i \exp[-jk_i \mathbf{n}_i \cdot \mathbf{r}] \quad (12)$$

where $\mathbf{n}_i$ is the two-dimensional vector

$$\mathbf{n}_i = -\sin \theta_i \mathbf{a}_y + \cos \theta_i \mathbf{a}_z \quad (13)$$

and where $\mathbf{r}$ denotes the usual position vector

$$\mathbf{r} = x\mathbf{a}_x + y\mathbf{a}_y + z\mathbf{a}_z. \quad (14)$$

Additionally, k_i is the (generally complex) propagation constant, related to the properties of the i th layer according to

$$k_i^2 = \omega^2 \mu_i \epsilon_i - j\omega \mu_i \sigma_i = \omega^2 \mu_i \epsilon_i (1 - j \tan \delta_i). \quad (15)$$

Without loss in generality, all observations can be made in the plane $y = 0$. Then, $\mathbf{n}_i \cdot \mathbf{r} = z \cos \theta_i$ and the wave traveling left to right becomes

$$A_i \exp[-jk_i z \cos \theta_i]. \quad (16)$$

Likewise, the wave traveling right to left can be written as

$$B_i \exp[jk_i z \cos \theta_i]. \quad (17)$$

Therefore, in terms of

$$\gamma_i = k_i d \cos \theta_i \quad (18)$$

the continuity of the electric field intensity vector at $z = d$ (see Fig. 5) can be expressed as

$$A_i \exp(-j\gamma_i) + B_i \exp(j\gamma_i) = A_{i+1} + B_{i+1}. \quad (19)$$

Equation (19) is consistent with [8, eq. (13)], which for the case of perpendicular incidence also gives

$$\begin{aligned} \cos \theta_i [A_i \exp(-j\gamma_i) - B_i \exp(j\gamma_i)] \\ = (k_{i+1}/k_i) \cos \theta_{i+1} (A_{i+1} - B_{i+1}) \end{aligned} \quad (20)$$

as an expression for continuity of the magnetic field intensity vector at $z = d$. It is important to note that the (generally complex) angle of refraction θ_i can be determined from a simple extension of Snell's law, namely,

$$k_1 \sin \theta_1 = k_2 \sin \theta_2 = \dots = k_i \sin \theta_i = \dots = k_{N+2} \sin \theta_{N+2} \quad (21)$$

where N is the total number of layers. As shown in Fig. 3, θ_1 is the angle of incidence on the first layer.

Now, the desired recursion formulas can be obtained by simultaneous solution of (19) and (20). The results, for perpendicular incidence, are

$$A_i = [\frac{1}{2} \exp(j\gamma_i)] [A_{i+1}(1 + Y_{i+1}) + B_{i+1}(1 - Y_{i+1})] \quad (22)$$

$$B_i = [\frac{1}{2} \exp(-j\gamma_i)] [A_{i+1}(1 - Y_{i+1}) + B_{i+1}(1 + Y_{i+1})] \quad (23)$$

where

$$Y_{i+1} = \frac{\cos \theta_{i+1}}{\cos \theta_i} \left(\frac{\kappa_{i+1}(1 - j \tan \delta_{i+1})}{\kappa_i(1 - j \tan \delta_i)} \right)^{1/2}. \quad (24)$$

In the region to the right of the multilayered slab there is only a wave traveling left to right, whose amplitude and phase can be set (arbitrarily) equal to one and zero, respectively. Therefore, setting $A_{N+2} = 1$ and $B_{N+2} = 0$ allows A_i and B_i to be calculated from (23) and (24) by setting $i = N + 1, N, N - 1, \dots, 3, 2, 1$, in that sequence. Obviously, A_1 is the incident wave on the left side of the slab, and B_1 is the reflected wave. As a result, the complex transmission coefficient for perpendicular incidence is simply

$$T_{\perp} = 1/A_1. \quad (25)$$

The analysis for parallel incidence follows a similar course, with but slight changes in notation, and requires use of [8, eq. (17)]. The results are

$$F_i = [\frac{1}{2} \exp(j\gamma_i)] [F_{i+1}(1 + W_{i+1}) + G_{i+1}(1 - W_{i+1})] \quad (26)$$

$$G_i = [\frac{1}{2} \exp(-j\gamma_i)] [F_{i+1}(1 - W_{i+1}) + G_{i+1}(1 + W_{i+1})] \quad (27)$$

where

$$W_{i+1} = \frac{\cos \theta_{i+1}}{\cos \theta_i} \left(\frac{\kappa_i(1 - j \tan \delta_i)}{\kappa_{i+1}(1 - j \tan \delta_{i+1})} \right)^{1/2} \quad (28)$$

$$T_{\parallel} = 1/F_1. \quad (29)$$

It is clear that both $T_{\perp}$ and $T_{\parallel}$ are (generally complex) scalar quantities.

Now, the tangential component of the electric field intensity vector, at a point on the outer surface of the radome just opposite (x', y', z') along the normal $\mathbf{n}_0$, is given by

$$\mathbf{E}_t = [(\mathbf{t} \cdot \mathbf{E}) \mathbf{t}] T_{\perp} + [\eta(\mathbf{t} \cdot \mathbf{H}) \mathbf{t} \times \mathbf{n}_0] T_{\parallel} \quad (30)$$

where $\eta = 120\pi$ ohms is the characteristic impedance of free space. The corresponding expression for tangential $\mathbf{H}$ is obtained from

$$\mathbf{H}_t = (1/\eta) \mathbf{n}_0 \times \mathbf{E}_t \quad (31)$$

and completes the information required for calculating the desired antenna pattern.

INTEGRATION

For this purpose, [7, eqs. (3) and (4)] are used, noting first that in both equations $(\mathbf{n} \times \mathbf{E})$ and $(\mathbf{n} \times \mathbf{H})$ are just the expressions defined by (30) and (31) of this work. Additionally, r denotes radial distance, as shown in Fig. 1, from a point on the radome surface to the point of observation, and $\mathbf{u}_r$ is a unit vector along the ray to (x_0, y_0, z_0) . It should be noted that *this* coordinate triplet plays the role of (x', y', z') in [7, eqs. (3) and (4)], and that integration is to be performed with respect to the primed coordinates in Fig. 1. In the final analysis, both integrations reduce to

TABLE II
COMPARISON BETWEEN FAR-FIELD AND NEAR-FIELD PATTERNS OF HORN ANTENNA (DECIBELS)

R_0 (meters)	θ									
	0°	10°	20°	30°	40°	50°	60°	70°	80°	90°
H Plane										
$R_0 = 0.3$	0	-0.75	-4.88	-11.45	-15.92	-22.84	-11.46	-8.81	-7.47	-7.95
$R_0 = 1000$	0	-5.91	-15.30	-23.95	-30.62	-37.92	-41.22	-44.18	-44.10	-46.10
E Plane										
$R_0 = 0.3$	0	-1.74	-4.15	-8.75	-14.99	-15.32	-22.44	-22.21	-21.22	-22.72
$R_0 = 1000$	0	-7.86	-10.80	-16.92	-25.26	-20.75	-30.82	-27.70	-26.24	-27.60

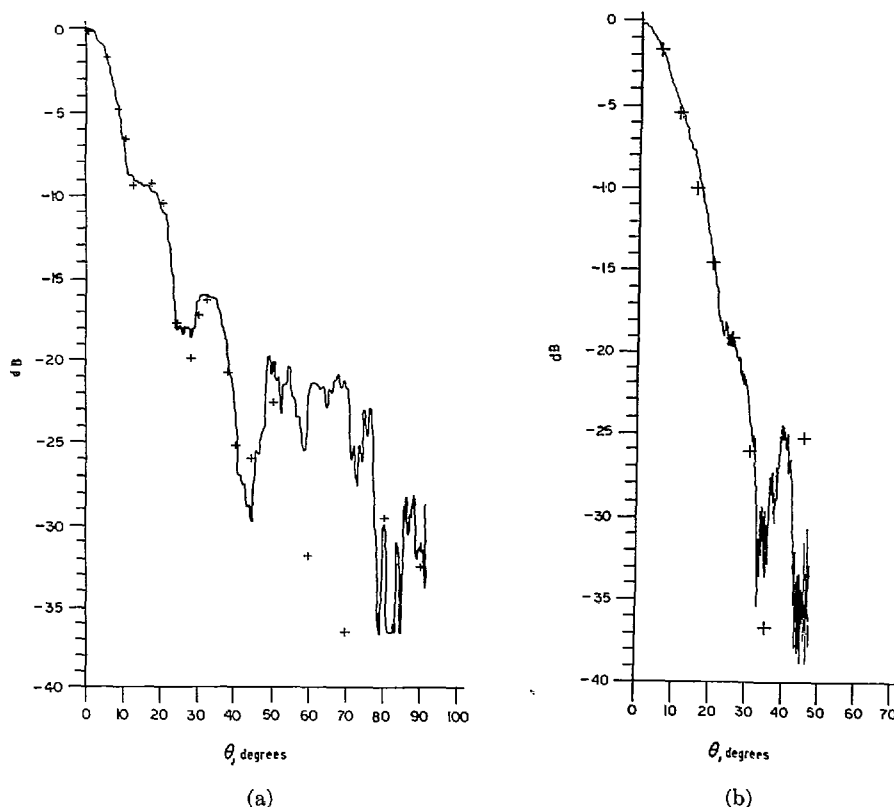


Fig. 6. Measured (solid) and calculated (crosses) patterns for $\Omega = 0^\circ$. (a) *E* plane. (b) *H* plane.

simple summations when they are performed on the computer.

Two extremely important questions must now be posed.

- 1) How large an area of the radome surface should be covered by the integration?
- 2) How far apart should the sample points be spaced?

Quite evidently, the answers to both questions hinge strongly on the nature of the field at all points on the radome surface. It is natural to expect that every point where the field is relatively large should be taken into account by the integration. In fact, since the near-field pattern of an antenna is generally broader than the far-field pattern, the range of the summation should be determined only after a study of the near field has been made.

Table II provides a comparison between the calculated far-field and near-field patterns of the horn antenna shown in Fig. 1, and shows a definite beam broadening effect in

the near field. (The far-field patterns were verified experimentally.) In this table R_0 is the radial distance from the point of observation to the center of the aperture, and θ is the polar angle measured from the z axis in Fig. 1. The distance $R_0 = 0.3$ meter is roughly the distance from the center of the aperture to points on the radome surface.

Inspection of Table II shows that the integration should span over a sufficiently large area so as to take into account every Huygens source whose contribution to the pattern might be significant. Accordingly, it is convenient to define an angle $\theta_{\max}$, measured with respect to the z axis in Fig. 1, as the angle at the apex of a cone whose intersection with the curved source aperture establishes the proper limits on the curved source aperture. Naturally, there are other criteria that can be used equally well, but the results reported here are couched in terms of the limiting angle $\theta_{\max}$.

Now, the separation between sample points on the surface subtended by this cone must be chosen small enough

TABLE III
VALUES AS $\theta_{\max}$ VARIES (DECIBELS)

$\theta_{\max}$	θ					
	0°	10°	20°	30°	40°	50°
<i>H</i> plane						
40°	0	-6.48	-17.64	-31.62	-46.26	-43.34
50°	0	-5.46	-15.21	-24.84	-26.32	-27.42
60°	0	-5.82	-14.65	-25.97	-40.06	-21.63
75°	0	-5.64	-14.85	-24.28	-35.76	-31.47
90°	0	-5.70	-15.11	-24.67	-30.74	-24.80
<i>E</i> Plane						
50°	0	-6.79	-9.60	-17.33	-31.50	-29.61
60°	0	-7.49	-10.39	-17.26	-25.11	-22.66

TABLE IV
PATTERN VARIATIONS DUE TO CHANGES IN SPACING (DECIBELS)

Spacing	θ										
	0°	5°	10°	15°	20°	25°	30°	35°	40°	45°	50°
<i>H</i> plane											
$\lambda/3$	0	-1.76	-5.82	-10.06	-14.65	-19.16	-25.97	-36.60	-40.06	-25.05	-21.63
$2\lambda/3$	0	-1.67	-5.71	-10.32	-14.32	-19.31	-26.83	-35.63	-47.95	-24.72	-21.16
λ	0	-1.62	-5.84	-10.49	-14.90	-18.47	-24.32	-34.78	-51.42	-23.91	-20.67
$4\lambda/3$	0	-1.75	-6.29	-10.82	-16.43	-17.33	-23.31	-24.58	-16.51	-8.49	-13.78
2λ	0	-2.01	-6.72	-9.74	-13.02	-5.23	-3.11	-9.10	-12.84	-21.08	-10.76
<i>E</i> plane											
$\lambda/3$	0	-1.75	-7.49	-9.28	-10.39	-20.47	-17.26	-17.31	-25.11	-24.51	-22.66
$2\lambda/3$	0	-1.66	-7.51	-9.22	-10.40	-22.10	-16.58	-16.76	-25.25	-23.05	-22.33
λ	0	-1.98	-6.80	-9.04	-8.87	-18.27	-20.71	-16.20	-24.65	-22.01	-22.91
$4\lambda/3$	0	-1.63	-6.84	-7.89	-7.66	-15.54	-20.23	-14.75	-21.01	-15.20	-23.56
2λ	0	-0.90	-9.80	-10.10	-11.41	-16.74	-29.66	-14.96	-11.12	-8.70	-8.42

so as to allow frequent sampling relative to a wavelength, yet large enough so as to keep computer time down to a minimum. The selection of the step size is based on the observation that the integrands in [7, eqs. (3) and (4)] contain an exponential—and therefore periodic—term, and hence they should be sampled more than once per cycle. This conclusion is confirmed by the results of this investigation on the basis of several step sizes used to perform the numerical integration. In this regard, each Huygens source is taken into account in proper sequence, with the relatively weak ones grouped together to complement the effects of stronger sources. For Ω small, one such sequence is to increment Y and to then go completely around the radome surface following a path such as shown in Fig. 4, and making sure that no two sample points are farther apart than the preselected step size. Of course, once the sample points are set, the corresponding differential areas on the radome surface can be calculated readily.

NUMERICAL RESULTS

As indicated in [7], the vector diffraction theory is capable of providing the magnitude and phase angle of all three spatial components of both $\mathbf{E}$ and $\mathbf{H}$ at any point (x_0, y_0, z_0) such that $r \geq 1.59\lambda$. For points in the far field, however, the wave is locally plane and the calculation of the power pattern can be based on the simple formula

$$\text{dB} = 10 \log_{10} \frac{\mathbf{E} \cdot \mathbf{E}^*}{[\mathbf{E} \cdot \mathbf{E}^*]_0} \quad (32)$$

where the subscript zero in the denominator indicates that

the value of $\mathbf{E} \cdot \mathbf{E}^*$ at $\theta = 0$ is to be taken as a reference. Clearly, only the $\mathbf{E}$ field is needed to calculate the desired power pattern.

Accordingly, Fig. 6 shows principal plane patterns for $\Omega = 0^\circ$, $\theta_{\max} = 60^\circ$, and for a step size of $\lambda/3$. From these figures, it can be seen that agreement between theory and experiment is good as far as the main beam is concerned, fairly satisfactory in the region of the principal sidelobes, and adequate near the noise levels. As $\theta_{\max}$ is allowed to vary, some variations are noted, especially beyond the main lobe, as shown by the comparison made in Table III. On the other hand, pattern variations due to changes in spacing between sample points are shown in Table IV for $\theta_{\max} = 60^\circ$. From this table it is apparent that basically similar answers are obtained as the sample point spacing is increased from $\lambda/3$ up to (even) λ , and that beyond a full wavelength the differences tend to become appreciable. This fact is very significant since computer time is understandably a function of sample point spacing, in addition to being a function of the extent to which the radome surface is covered by the integration. Specifically, for $\theta_{\max} = 60^\circ$ and a sample point spacing of $2\lambda/3$, the time taken by the Univac 1108 computer to calculate the tangential field on the exterior radome surface was 310 seconds; and it took 106 additional seconds to generate, by integration, 14 points on each principal plane pattern. The corresponding figures for $\theta_{\max} = 90^\circ$ and a spacing of $\lambda/3$ were 40 minutes and 426 seconds, respectively. However, it is not necessary to go to the extremes implied by the second case in order to obtain reasonably acceptable results.

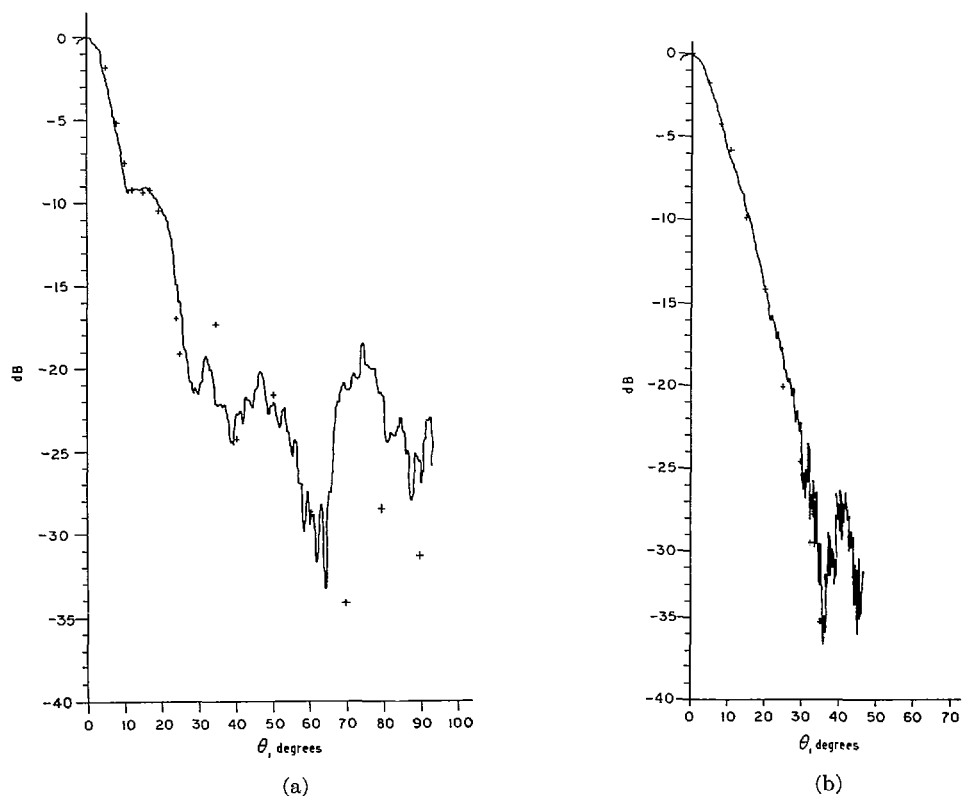


Fig. 7. Measured (solid) and calculated (crosses) patterns for $\Omega = 15^\circ$. (a) E plane. (b) H plane.

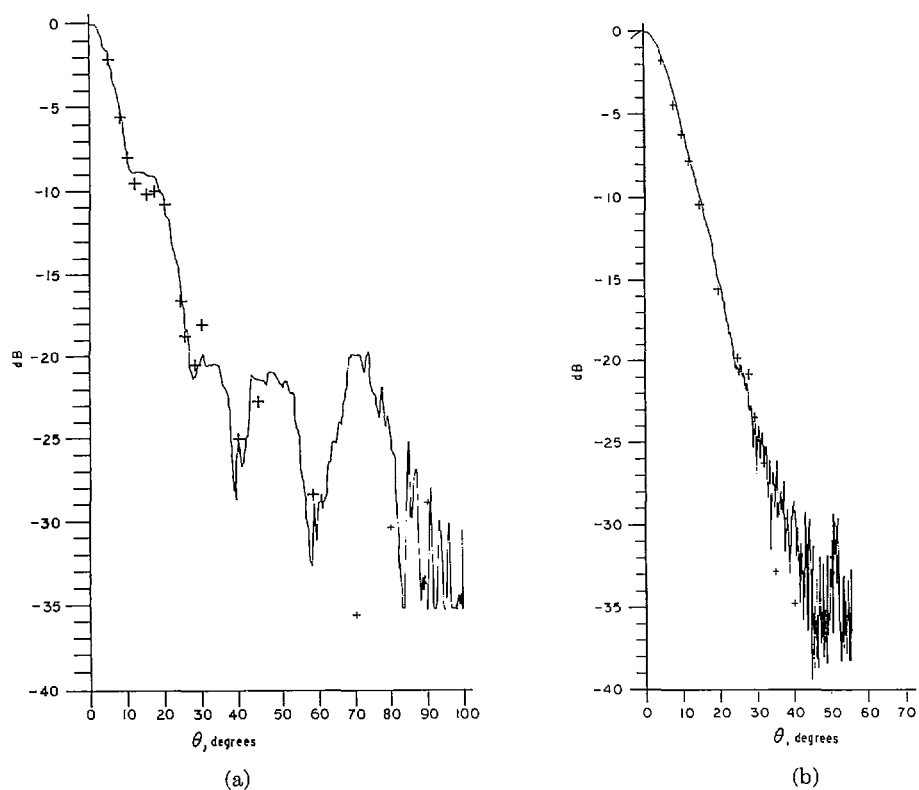


Fig. 8. Measured (solid) and calculated (crosses) patterns for $\Omega = 30^\circ$. (a) E plane. (b) H plane.

It should be observed that, in addition to the principal plane patterns, other patterns, or cuts, can be obtained just as easily once the tangential field is known over the radome surface.

Finally, Figs. 7 and 8 show calculated and measured patterns for $\Omega = 15^\circ$ and $\Omega = 30^\circ$, respectively. In both cases $\theta_{\max} = 60^\circ$ and the spacing is $2\lambda/3$. Good agreement is to be noted here, too, between measured and calculated patterns.

Although the results presented here are basically encouraging, no general conclusions can be drawn from a single case study of this sort. It was mainly the intent of this presentation to outline a computational procedure and not to attempt to answer, in general, the question of sample point spacing and extent of surface integration.

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Mutual Coupling Between Log-Periodic Antennas

ROBERT H. KYLE, MEMBER, IEEE

Abstract—Numerical techniques are used for the analysis of mutual coupling in arrays of log-periodic dipole antennas. The analysis is formulated in terms of impedance and admittance matrices for the dipole and transmission line networks. Approximations are made to allow the solution for moderately sized arrays to be practical with the existing generation of computers. Antenna patterns for the array are expressed in terms of the current incident on each log-periodic antenna. Results are presented for systems of one, two, and three log-periodic antennas.

I. INTRODUCTION

THE log-periodic (LP) dipole antenna has certainly been one of the most useful of the frequency independent antennas developed in recent years. One significant advantage of these antennas is the straightforward design procedure [1] which facilitates their engineering application. It is not surprising that LP antennas are now being

postulated as elements of an array. This has generated an interest in their mutual coupling properties, the subject of the present paper.

The problem is attacked through a basically numerical approach. The application of such techniques to electromagnetics problems has been developing at a rapid pace recently. Numerical techniques make possible the solution of a large number of electromagnetics problems which, by classical techniques, would be virtually impossible. However, the solution of a problem involving a body many wavelengths in extent can be extremely difficult even with numerical techniques. The limitation is not a theoretical one. A set of equations describing the system can be generated, but the set is simply too large to be solved in a practical way. In general, the size of the set of equations which can be dealt with is limited by the storage capacity, speed, and to a lesser extent, the round-off error of currently available computers. In the analysis to follow, two basic approximations are made.

1) Current on the individual dipoles is assumed to be segmented. The complex amplitude of each segment is determined by point matching the electric field.

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The author was with the General Electric Company, Syracuse, N. Y. He is now with Radiation Systems, Inc., McLean, Va.